DATA WAREHOUSING AND DATA MINING

(Common to CSE, IT and AI&DS)

III Year – I Semester

Lecture : 3	Internal Marks : 30
Credits : 3	External Marks : 70

==========

Course Objectives

- To introduce basic concepts and techniques of data warehousing and data mining.
- To examine the types of the data to be mined and apply pre-processing methods on raw data.
- To discover interesting patterns, analyze supervised and unsupervised models and estimate the accuracy of the algorithms.

Course Outcomes

Upon successful completion of the course, the students will be able to

- describe core data warehouse concepts and the importance of data mining.
- use data preprocessing techniques on raw data to make it appropriate for mining.
- apply classification algorithms to solve classification problems.
- apply association rule mining to identify frequent item sets in large databases.
- evaluate different clustering techniques to determine their appropriateness for the given dataset.

Course Content

UNIT – I: Data Warehousing and Online Analytical Processing

Basic concepts, Data Warehouse Modeling: Data Cube and OLAP, Data Warehouse Design and Usage, Data Warehouse Implementation, Cloud Data Warehouse, Data Mining and Patten Mining, Technologies, Applications, Major issues, Data Objects & Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity. (Text Book- 1)

UNIT – II: Data Preprocessing

An Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization. (Text Book- 1)

UNIT – III: Classification

Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction, Visual Mining for Decision Tree Induction, Bayesian Classification Methods: Bayes Theorem, Naïve Bayes Classification, Rule-Based Classification, Model Evaluation and Selection. (Text Book- 2)

UNIT – IV: Association Analysis

Problem Definition, Frequent Itemset Generation, Rule Generation: Confident Based Pruning, Rule Generation in Apriori Algorithm, Compact Representation of frequent item sets, FP-Growth Algorithm. (Text Book- 2)

UNIT – V: Cluster Analysis

Overview, Basics and Importance of Cluster Analysis, Clustering techniques, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bi-secting K Means, Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses. (Text Book- 2)